



Universidad Autónoma del Estado de Hidalgo

Instituto de Ciencias Básicas e Ingeniería

School of Engineering and Basic Sciences

Área Académica de Matemáticas y Física

Mineral de la Reforma, Hgo., a 25 de agosto de 2026

Número de control: ICBI-AAMyF/2843/2026

Asunto: Letter of Support for Dr. Philip J. Gerrish and Professor Gabriela Gomes

Resistance Networks Selection Committee

Wellcome Leap

Re: Letter of Support for Dr. Philip J. Gerrish and Professor Gabriela Gomes

Proposal: "Universal Properties of Plasmid Evolution and Robust Risk Forecasting for Preemptive pARG Intervention"

Dear Members of the Selection Committee:

We are pleased to express our strong support for the proposal entitled "Universal Properties of Plasmid Evolution and Robust Risk Forecasting for Preemptive pARG Intervention," submitted to the Wellcome Leap Resistance Networks program. In particular, we support Dr. Philip J. Gerrish's proposed leadership of the initiative's mathematical and evolutionary-modeling efforts and enthusiastically welcome the opportunity to collaborate with both Dr. Gerrish and the proposed co-Principal Investigator, Professor Gabriela Gomes. Dr. Gerrish has established a substantial and productive intellectual presence within the Mathematics Department at the Universidad Autónoma del Estado de Hidalgo (UAEH). He has worked closely with our graduate students, directed and contributed to graduate thesis research, and collaborated with faculty members on problems at the intersection of mathematics, evolutionary biology, and complex systems. These projects have addressed fundamental questions in evolutionary theory while developing applications to ecology, epidemiology, and cancer. Through this work, we have seen firsthand his ability to formulate biologically important problems in mathematically precise terms, develop tractable theoretical frameworks, and guide students and collaborators in connecting abstract results to empirical questions.

The proposed Resistance Networks project represents a particularly compelling application of these strengths. The spread of antibiotic-resistance plasmids is not solely a microbiological problem; it is an evolutionary and dynamical problem involving interactions among bacterial hosts, mobile genetic elements, environmental conditions, and transmission networks across multiple biological and temporal scales. Progress therefore requires models that can integrate processes operating within bacterial populations, within individual hosts, and across communities and environments. Dr. Gerrish's research on the evolutionary consequences of horizontal genetic transfer, recombination, selection, and linkage provides a strong conceptual and mathematical foundation for this work.

"Amor, Orden y Progreso"



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We believe that Dr. Gerrish is especially well suited to lead the modeling component of a large interdisciplinary initiative of this kind. His approach combines fundamental evolutionary theory with explicit, falsifiable predictions that can be tested using genomic, epidemiological, and longitudinal population data. Equally important, he understands modeling as a collaborative and iterative activity: models should help determine which measurements are most informative, while empirical results should continually refine the models and their assumptions. This capacity to connect theoretical development with experimental and observational work will be essential to the success of the Resistance Networks program.

We are also very enthusiastic about Professor Gabriela Gomes's participation as co-Principal Investigator. Her expertise in mathematical epidemiology, population heterogeneity, and the dynamics of infectious disease complements Dr. Gerrish's expertise in evolutionary theory and population genetics exceptionally well. Their combined perspectives provide a powerful basis for linking the evolution of plasmids within bacterial populations to patterns of transmission and risk at broader epidemiological scales. We believe that this partnership will strengthen both the theoretical foundations and the practical forecasting objectives of the proposed research.

Dr. Gerrish plans to involve graduate students with whom he is already working at UAEH in appropriate aspects of the project. These students have relevant mathematical training and are already supported through existing funding arrangements. Their participation would provide the project with additional modeling capacity while giving them an exceptional opportunity to contribute to an internationally significant problem in antimicrobial resistance. Faculty members and graduate students at UAEH would be excited to collaborate with both Dr. Gerrish and Professor Gomes, to contribute our mathematical expertise to the project, and to participate in the broader exchange of ideas, methods, and training opportunities that this collaboration would create.

Our experience working with Dr. Gerrish gives us great confidence in both his scientific judgment and his ability to mentor young researchers and sustain productive interdisciplinary collaborations. Together, Dr. Gerrish and Professor Gomes bring an unusual and highly valuable combination of mathematical depth, evolutionary insight, epidemiological expertise, creativity, and commitment to addressing consequential biological problems. We expect that their leadership would make the modeling effort scientifically rigorous, conceptually innovative, and closely integrated with the broader empirical and translational goals of Resistance Networks.



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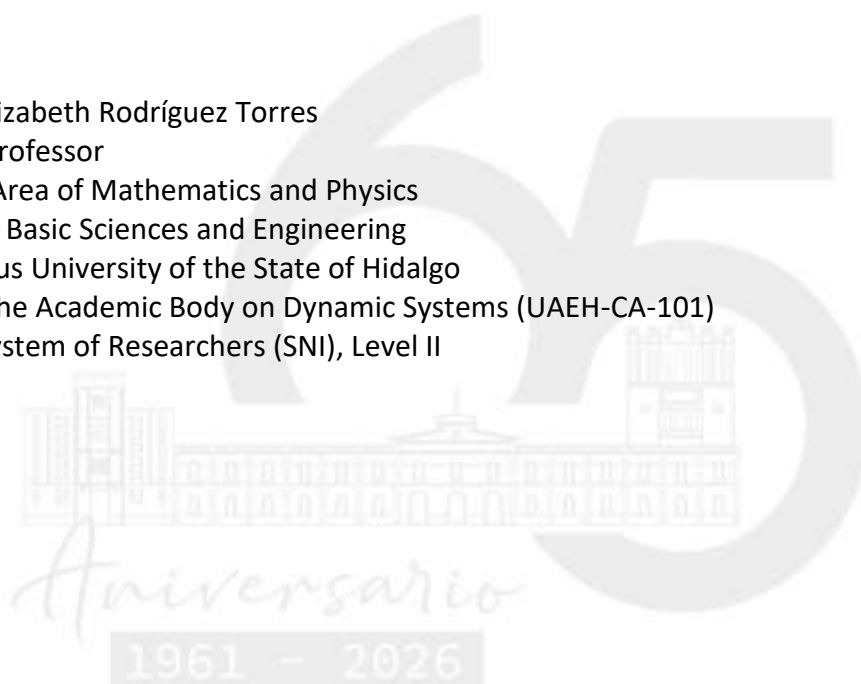
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We therefore offer our enthusiastic support for this proposal. Subject to the appropriate institutional arrangements, we would welcome the participation of our faculty and graduate students in this research and look forward to collaborating with Dr. Gerrish, Professor Gomes, and the other teams participating in the Resistance Networks program. Please do not hesitate to contact us if further information would be helpful.

Sincerely,

Dr. Erika Elizabeth Rodríguez Torres
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